Qwen

A Governance Framework for BCI-Integrated Mixed-Reality Horror Systems

Three-Tier Architecture: Separating Research Governance from Engine Implementation

The foundational principle underpinning the development of Horror$Place's vm-constellation is a strict three-tier governance architecture, which creates a robust separation of concerns between high-level research and standardization (Tier 3) and low-level runtime implementation (Tier 2) . This architectural decision is critical for managing the inherent complexity and rapid evolution of Brain-Computer Interface (BCI) technology while ensuring the stability and performance demanded by a commercial-grade game engine like Unreal or Unity. The distinction between these tiers maps cleanly onto "lab/research tooling + constellation governance" (Tier 3) and "engine/runtime implementation" (Tier 2) . This separation allows researchers to innovate on core algorithms—such as fear detection models and adaptive calibration protocols—in a sandboxed environment without risking the integrity of the production codebase [52] . Conversely, developers can build robust, performant engine integrations based on stable, well-defined data contracts, insulated from the fluid nature of BCI signal processing science [6] .

Tier 2, the engine-level implementation layer, is tasked with consuming the outputs of the Tier 3 governance structures. In this paradigm, the BCI is not treated as a raw, unprocessed EEG stream but as a schematized source of metrics that modulates gameplay systems like the `HorrorDirector` . The guiding pattern for Tier 2 implementation is to keep engine-facing APIs minimal and abstract away all complex numerical computations. All numerics, including Exponentially Moving Average (EMA) smoothing, feature extraction pipelines, and metric mappings, are handled within a dedicated Rust library . The engine plugins for Unity and Unreal act as thin, safe wrappers around this core Rust logic, exposing only a narrow surface area for consumption. This approach is heavily influenced by best practices in distributed systems and cloud-native application design, where clear interfaces and service-level objectives (SLOs) are paramount for scalability and reliability [23] [52] . By enforcing this contract-first approach, the system ensures that any changes to the underlying BCI processing pipeline can be deployed independently of the game engines themselves, reducing coupling and increasing development velocity.

Tier 3 represents the research and governance nexus of the entire system. It is responsible for defining the rules of engagement that all other tiers must follow. This includes finalizing and extending canonical data schemas such as `bci-feature-envelope` and `bci-metrics-envelope`, building a Rust Command Line Interface (CLI) for ingesting and converting external BCI datasets, and implementing the core Rust mapping engine crate . Furthermore, Tier 3 is where the policies for human-AI co-development are established through documents like the `BCI-Dev-Guide` and the `AI-BCI-PLAYBOOK` . These artifacts provide non-negotiable rules for how BCI-related features are designed, implemented, and validated. The explicit directive to prioritize Tier 3 governance over Tier 2 implementation gaps reinforces this hierarchy: the system must be built upon its own contracts, not the other way around . This tier also houses the lab-to-engine feedback mechanisms, such as synthetic trace libraries and NDJSON replay tools, which are essential for refining the system's behavior based on empirical data gathered during controlled experiments . The entire vm-constellation operates under these Tier 3 contracts, ensuring that all components—from a Unity prefab to a Rust crate—conform to a single, coherent set of standards. This governance model is analogous to a formal reference architecture for cloud-native applications, which streamlines governance across a complex ecosystem of services and provides a sample implementation for clarity [52] . It establishes a clear boundary between innovation and production, allowing the project to evolve scientifically while maintaining a solid engineering foundation.

Adaptive Calibration Design for AR Horror Contexts

The design of the adaptive calibration protocol is central to the success of the BCI-AR horror system, directly addressing the research goal's emphasis on user-state modeling and fear-detection thresholds over raw performance optimization . The objective is to create a short, effective calibration process that takes less than two minutes while achieving greater than 90% accuracy in detecting user states relevant to horror mechanics, such as fear . The protocol is fundamentally per-user, acknowledging neurodiversity by learning individual baselines and thresholds rather than relying on a one-size-fits-all model. This personalization is achieved through a multi-phase mini-game sequence executed within a dedicated calibration scene in either Unity or Unreal .

The calibration process consists of distinct epochs, each designed to elicit a specific physiological state. These epochs are labeled in the telemetry metadata, allowing the backend processing to correctly interpret the user's state during each phase . The first epoch, `CALM_BASELINE`, typically involves a 30–60 second period where the user is

instructed to relax, often through breathing exercises in an idle VR room . This establishes a baseline of calm brain activity. The second epoch, `FOCUSED_TASK`, presents a mild cognitive challenge, such as mental arithmetic or target tracking, to induce a state of focused attention . The third, and particularly critical for AR, is the `AR_ALIGNMENT` phase, which involves gaze-locked AR targets to calibrate the user's spatial awareness and ensure virtual elements are properly anchored to their real-world environment . The data collected during these epochs—the minimum and maximum values for key features, artifact thresholds, and optimal EMA smoothing parameters—are processed by a dedicated BrainFlow feature server running a canonical pipeline . This server applies filters, computes power spectral density (PSD), extracts bandpower features, and calculates composite stress and attention indices before emitting them as standardized feature envelopes . The engine itself is never involved in this computationally intensive processing; its role is passive reception .

The results of this lab-side analysis are stored in external, machine-readable configuration files, such as JSON or TOML, which are consumed by the engine at runtime . These calibration profiles contain all the necessary information for the Rust mapping engine to normalize incoming BCI data to a consistent 0–1 scale for each individual user. For example, the profile might specify the min/max values for `raw_attention` and `raw_startle` and the EMA alpha value for smoothing each channel . This decoupling of calibration from runtime execution is a cornerstone of the Tier 3 governance model. It allows researchers to continuously refine the calibration algorithm and generate better-performing profiles without requiring a new build of the game engine. The engine simply loads the latest calibration file and proceeds with its predefined logic . To further enhance safety and user experience, the `HorrorDirector` subsystem implements guardrails that monitor signal quality and user state. If the signal degrades or if the user's stress score remains above a certain threshold for too long, the director can clamp tension, enforce cooldown periods, or temporarily disable BCI influence, falling back to pre-scripted pacing curves . This DET-aware safety mechanism is crucial for preventing user overload, especially in a mixed-reality context where disorientation could pose physical risks [51] .

Schema Enforcement and NDJSON Telemetry as Constellation-Wide Contracts

The bedrock of the Tier 3 governance structure is a rigorous enforcement of data schemas and the use of Newline Delimited JSON (NDJSON) for all telemetry. This practice serves

as the "contract card" that ensures consistency, interoperability, and type safety across the entire vm-constellation, spanning multiple repositories and programming languages . Instead of relying on ad-hoc data structures or raw waveforms, every piece of BCI-related data must conform to a strongly-typed, canonical schema . Key schemas include `EEGFeatureContract/BCIFeature` for raw and processed BCI features, and `bci-metrics-envelope` for the final derived entertainment metrics like UEC (Unpleasant Emotional Content) and ARR (Anxiety Response Rate) . These schemas are not merely descriptive; they are prescriptive, defining the exact fields, data types, and canonical ranges for all information exchanged, including vital contextual metadata such as CIC (Contextual Invariant Count), AOS (Active Objective State), DET (Dread Event Tension), and SHCI (Scare Hit Count Index) . This schema-first approach is a direct response to the challenges of managing complex, distributed systems where data integrity is paramount [2] [34]. By mandating adherence to these contracts, the system prevents common errors like malformed payloads, missing fields, or incorrect data types, which can otherwise lead to unpredictable behavior and difficult-to-debug issues.

NDJSON is chosen as the universal telemetry format because it offers an ideal balance between efficiency and readability. Each line in an NDJSON log file represents a single, self-contained data point—a snapshot of the system state at a specific time window . A typical entry in the telemetry log would contain a complete object with the BCI feature envelope, the corresponding derived metrics envelope, a full set of invariants, and the entertainment metrics for that frame . This rich, structured logging is indispensable for the lab-to-engine feedback loops. It creates a ground truth dataset that can be used to train and validate calibration models, tune mapping functions, and analyze the effectiveness of different horror scenarios . The ability to deserialize this JSON directly into typed structs (e.g., C++ `USTRUCT`s in Unreal) is a powerful enabler of this process, as it eliminates the error-prone manual parsing of key-value pairs and leverages the compiler for type safety [42]. For instance, Unreal's `FJsonObjectConverter::JsonObjectStringToUStruct` function can recursively populate a nested struct hierarchy from a JSON string, making data processing straightforward and reliable [42].

Validation is not an afterthought but an integral part of the governance workflow, enforced through the Continuous Integration (CI) system. Any NDJSON file submitted as part of a pull request is automatically validated against the official schemas before being merged . This acts as a gatekeeper, immediately rejecting any deviations from the established contract. This process is analogous to an Artificial Intelligence Driven Compliance Checker Engine (AICCE) used for verifying network protocol compliance, where correctness is verified programmatically [3]. Furthermore, the CI system validates

Lua scripts to ensure they only invoke approved BCI Application Programming Interfaces (APIs) exposed by the Rust C ABI, preventing unauthorized or unsafe operations . This closed-loop validation system ensures that the integrity of the entire vm-constellation's data contract is maintained. The prohibition of raw waveforms and Personally Identifiable Information (PII) in these logs is strictly enforced, with banned artifacts moved to secure staging areas for processing, ensuring user privacy and security [89] [90] . This disciplined approach to schema and telemetry management provides a scalable and auditable foundation for both current development and future expansion of the BCI-integrated system.

Schema Component	Description	Key Fields	Enforcement Mechanism
`bci-feature-envelope`	Raw and processed BCI signals from a single time window.	`timestamp`, `device_id`, `features`: `{attention, startle, breath, ...}`, `metadata`: `{epoch_type, signal_quality}`	NDJSON validation against JSON Schema; `serde_json`'s `deny_unknown_fields` attribute [89] .
`bci-metrics-envelope`	Derived entertainment metrics calculated from feature envelopes.	`timestamp`, `metrics`: `{UEC, EMD, STCI, CDL, ARR, ...}`, `context`: `{CIC, AOS, DET, ...}`	NDJSON validation against JSON Schema; deserialization into typed engine structs (e.g., Unreal `USTRUCT`) [42] .
`calibration-profile`	Per-user normalization and smoothing parameters for the BCI pipeline.	`user_id`, `device_model`, `calibration_version`, `parameters`: `[{"channel": "...", "min": x, "max": y, "ema_alpha": z}, ...]`	File validation against JSON Schema; loaded at engine startup via the Rust C ABI .
`experiment-config`	Definition of a controlled calibration or testing experiment.	`experiment_id`, `conditions`: `[{"name": "...", "params": {...}}]`, `blocks`: `[{"type": "...", "duration": ...}]`	Python/Lua orchestration scripts validate the config structure before launching an experiment run [1] .

AI-Playbook Compliance and Co-Development Governance

To safely harness the potential of AI in the development process, the vm-constellation employs a sophisticated governance framework centered on the `AI-BCI-PLAYBOOK`. This document acts as a constitution for AI developers, meticulously detailing the rules of engagement for creating, modifying, and analyzing BCI-related code and assets . Its primary goal is to ensure that AI-generated contributions align with the project's core tenets: schema-first design, strict API usage, and adherence to established contracts. This playbook is a practical application of alignment faking, a concept where an AI model is trained to selectively comply with training objectives when it infers that it is being monitored, thereby preventing it from inventing new, ungoverned logic [31] . The playbook

instructs the AI on how to perform tasks ranging from writing device wrappers for new EEG hardware to designing novel BCI-to-metric mappings and analyzing experimental telemetry logs . Crucially, it mandates that the AI always targets existing schemas and APIs, reinforcing the separation of concerns enforced by the three-tier architecture.

Before generating any code, the AI is required to perform a discovery phase, querying a centralized `schema-spine-index.json` file . This index acts as a manifest of all available schemas, NDJSON registries, and APIs across the various repositories in the HorrorPlace constellation, such as `HorrorPlace-Constellation-Contracts`, `Death-Engine`, and `Spectral-Foundry` . This step ensures the AI has a coherent and up-to-date map of the system it is operating within, preventing it from making changes in isolation or attempting to modify components outside its designated scope. This discovery mechanism is analogous to how modern Integrated Development Environments (IDEs) and Language Server Protocols (LSPs) provide context-aware suggestions and refactoring capabilities, but applied at the level of a distributed codebase [35] . By grounding the AI's actions in a known set of contracts, the system maintains its structural integrity and prevents the introduction of bugs that could arise from a lack of system-wide awareness.

To maintain control, auditability, and modularity, all AI-generated changes are constrained by two key mechanisms: prism envelopes and one-file-per-turn authoring . Prism envelopes encapsulate each AI-generated modification within a machine-checkable change description. This wrapper contains metadata such as the correct schema ID, repository path, and associated contract IDs, providing a clear audit trail for every change . This practice echoes principles found in blockchain-enabled systems, where transactions are immutable and verifiable, promoting trust and accountability [22] . The one-file-per-turn policy forces the AI to produce a discrete, self-contained file for each action, preventing sprawling, uncoordinated changes that are difficult to review and integrate . This modular workflow aligns with modern software development practices that favor small, incremental, and testable commits. Together, these mechanisms create a tightly controlled yet highly productive co-development environment. Humans and AI can collaborate effectively, with the AI acting as a powerful assistant that operates within a well-defined framework of rules and constraints, ultimately accelerating innovation without compromising the stability and quality of the final product [15] .

Lab-to-Engine Feedback Loops for System Refinement

The ultimate value of the Tier 3 governance framework is its capacity to establish continuous, data-driven feedback loops that inform and improve both the immediate engine integration and the long-term calibration of the BCI-AR horror system . This transforms the development process from a purely speculative endeavor into a scientific one, where hypotheses about player experience can be tested, measured, and refined based on empirical evidence. The primary vehicle for this feedback is the rich, structured NDJSON telemetry generated during controlled experiments conducted in the lab . These experiments involve running players through carefully designed scenarios with randomized scenes, comparing conditions such as `EEG-adaptive` versus `scripted` horror pacing to isolate the effect of the BCI system .

From these lab sessions, teams construct synthetic trace libraries and NDJSON replay tools . These tools allow developers to simulate a wide range of user scenarios and emotional responses in a deterministic, repeatable manner. This is invaluable for debugging engine behavior, optimizing performance-critical paths like the Rust FFI calls, and validating that safety overrides (like those tied to Dread Event Tension or DET) are functioning as intended . This technique is akin to using frame-accurate sampling engines to profile hardware metrics, providing deep insights into system performance under specific, reproducible load conditions [25]. By replaying logged data, the team can observe exactly how the engine responded to a given BCI input and make targeted adjustments to the mapping logic or calibration profiles.

The most significant refinement comes from correlating BCI-derived metrics with entertainment metrics. The NDJSON logs capture not only the BCI signals (`raw_attention`, `raw_stress`) but also the resulting entertainment metrics (`UEC`, `EMD`, `ARR`) and contextual invariants (`CIC`, `DET`) . This rich dataset enables researchers to move beyond simple fear detection and into the nuanced territory of emotional storytelling. By analyzing these correlations, they can tune the calibration curves and mapping families to achieve specific design goals. For example, they can identify the precise BCI stress levels that correspond to peak engagement (high ARR) without crossing into overload, thus maximizing the perceived scariness of a scenario while keeping the player comfortable and willing to continue playing . This process effectively translates subjective feelings of "fear" and "tension" into objective, measurable data points that can be optimized algorithmically. The goal is to find the sweet spot where the system achieves >90% accuracy in fear detection while consistently producing desired entertainment outcomes, turning the horror mechanic into a finely calibrated instrument for eliciting emotion [7].

This analytical approach extends to the integration with the dice-driven procedural content generation (PCG) system. The NDJSON telemetry schema is extended to log the outcomes of the dice rolls, the resulting dungeon graphs (generated via BSP/Voronoi techniques), trap placements, scare timings, and branching fate outcomes alongside the BCI and entertainment metrics [77]. This allows for deep analysis into which PCG patterns are most effective at eliciting fear under different BCI states. For instance, the team could discover that a particular combination of a maze-like dungeon layout, triggered during a moment of high BCI-calculated dread, leads to a statistically significant spike in ARR. This insight would confirm the narrative and mechanical synergy of that specific PCG pattern, transforming the dice engine from a simple source of randomness into a sophisticated tool for data-driven emotional design. The AI chatbot, guided by the `AI-BCI-PLAYBOOK`, can be taught to treat the dice engine as a contract-driven system, where BCI just biases the selection of outcomes within predefined constraints, rather than inventing new, ungoverned logic . This closed-loop system, where lab experiments generate data that is used to refine both the BCI calibration and the PCG logic, which in turn informs the design of future experiments, creates a powerful cycle of continuous improvement.

Concrete Implementations and Interlocking Artifacts

The theoretical framework of the three-tier architecture and Tier 3 governance is realized through a set of concrete, GitHub-ready artifacts and code stubs that define the interlocking contracts between the lab, the core numerics, and the engine implementations. These tangible objects serve as the bridge between the high-level design and the day-to-day development work, ensuring that all components adhere to the established standards. The provided materials detail several key artifacts, including a `BCI AR Calibration Toolkit` documentation and concrete C# and C++ stub files for Unity and Unreal, respectively [13]. These artifacts are not standalone solutions but are deeply integrated into the broader system, reusing existing BCI envelopes, invariants, and entertainment metrics defined in the core contracts .

The `BCI AR Calibration Toolkit` document serves as the master directory and integration contract for the entire system . It specifies the exact location of all relevant files within the `HorrorPlace-Codebase-of-Of-Death` repository, from the Rust C ABI exports in the `core/bci-mapping-engine` crate to the Unity plugin modules in `engines/unity/BCIARCalibration` and the Unreal module in `engines/unreal/BCIARCalibration` . It explicitly defines the responsibilities of each component,

emphasizing that the Rust engine is engine-agnostic, while the engine-specific plugins act as wrappers . This document clarifies the distinction between Tier 2 (profiles, experiment configs) and Tier 3 (prefabs, bindings, runtime scripts), outlining the flow of artifacts between these tiers . For example, it details how the calibration JSON files produced by Tier 2 research are consumed by the Tier 3 engine plugins at runtime to configure the mapping process .

The concrete C# and C++ stubs provide the practical implementation of the engine-facing side of this contract. The Unity C# script, `BCI_ARCalibrationController.cs`, is a MonoBehaviour that demonstrates how to interact with the Rust library via Platform Invocation Services (P/Invoke) . It defines a `struct RtpcFrame` that mirrors the C-compatible output structure from the Rust engine and uses `DllImport` to import the necessary functions, `bciloadcalibration` and `bcitickinputsample` . The script handles loading the calibration profile from a specified path, clamps debug inputs, and on each update, constructs an input sample from engine-provided invariants and metrics, then calls the Rust function to get the resulting audio RTPCs (Real-Time Parameter Controls) . Similarly, the Unreal C++ component, `UBCIARCalibrationComponent`, provides a Blueprint-friendly interface. It uses a `USTRUCT` named `FBCIRtpcFrame` to represent the output data and wraps the C ABI calls in a class that can be easily added to an actor in the editor . The header file declares the properties and functions exposed to Blueprints, while the source file contains the implementation of `TickComponent` and `LoadCalibration`, which handle path resolution and the actual call to the native Rust functions . Both stubs exemplify the principle of minimal, clean surfaces: they expose only what is needed to drive the AR effects, delegate all the heavy number-crunching to the Rust library, and remain agnostic to the specifics of the underlying BCI hardware or signal processing pipeline. They are designed to be dropped into the project and wired up to existing audio and visual systems, demonstrating a clear path for immediate engine integration that fully respects the Tier 3 governance contracts [28] .

Reference

1. Arxiv今日论文| 2026-03-12 - 闲记算法 http://lonepatient.top/2026/03/12/arxiv_papers_2026-03-12.html

2. Forensic Engineer's Logical Framework for Deterministic Truth https://www.linkedin.com/posts/maurice-lyles-39363215_logicrail-sigma16-rk4-activity-7440554712774463488-vyar
3. Computer Science - arXiv https://arxiv.org/list/cs/new
4. (PDF) Bridging Signal Intelligence and Clinical Insight - ResearchGate https://www.researchgate.net/publication/397569139_Bridging_Signal_Intelligence_and_Clinical_Insight_A_Comprehensive_Review_of_Feature_Engineering_Model_Interpretability_and_Machine_Learning_in_Biomedical_Signal_Analysis
5. Sensors, Volume 24, Issue 3 (February-1 2024) – 328 articles - MDPI https://www.mdpi.com/1424-8220/24/3?os72BPAaEjhgg=BIROcWzE4
6. System Design for Distributed Applications | PDF | Cache (Computing) https://www.scribd.com/document/974279475/Expian-System-assigmet
7. Pleasure systems in the brain - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC4425246/
8. [PDF] Exploring the Collective Unconscious in the Age of Digital Media https://www.researchgate.net/profile/Stephen-Schafer-3/publication/306319679_The_Dream_Analog_Psychecology_Video_Games/links/5f84785a458515b7cf7a31ba/The-Dream-Analog-Psychecology-Video-Games.pdf
9. 333333 23135851162 the 13151942776 of 12997637966 https://www.cs.princeton.edu/courses/archive/spring18/cos226/assignments/autocomplete/testing/words-333333.txt
10. Creating a global knowledge system for food security https://unesdoc.unesco.org/ark:/48223/pf0000120905
11. Toward Understanding the Design of Intertwined Human–Computer ... https://dl.acm.org/doi/10.1145/3590766
12. Astronomical Motifs and Sources in the Writings of H. P. Lovecraft https://www.jstor.org/stable/26868370
13. EEG-Based Brain-Computer Interactions in Immersive Virtual and ... https://www.researchgate.net/publication/370611758_EEG-Based_Brain-Computer_Interactions_in_Immersive_Virtual_and_Augmented_Reality_A_Systematic_Review
14. Complex, Intelligent and Software Intensive Systems - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-031-96096-3.pdf
15. Learning and Collaboration Technologies - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-031-05675-8.pdf
16. Media Space 20+ Years of Mediated Life - Springer Nature https://link.springer.com/content/pdf/10.1007/978-1-84882-483-6.pdf

17. Ude My For Business Course List | PDF | Microsoft Azure - Scribd https://www.scribd.com/document/1001052628/Ude-My-for-Business-Course-List
18. (PDF) Cloud Computing and Security - Academia.edu https://www.academia.edu/10847483/Cloud_Computing_and_Security
19. [PDF] 写在《科技之魅》的前面 - World Internet Conference https://www.wicinternet.org/pdf/WorldLeadingInternetScientificandTechnologicalAchievements20231101.pdf
20. [PDF] 北京邮电大学科技论文，专著（译著）统计2017 年度 https://kyy.bupt.edu.cn/__local/8/7C/A5/2FDC88F33CFA6DD5CE78850907F_CF3B77FF_8545C1.pdf?e=.pdf
21. Machine Learning - arXiv https://arxiv.org/list/cs.LG/new
22. Blockchain-Enabled Internet of Vehicles Applications - MDPI https://www.mdpi.com/2079-9292/12/6/1335
23. SLOC: Service Level Objectives for Next Generation Cloud Computing https://www.researchgate.net/publication/343201327_SLOC_Service_Level_Objectives_for_Next_Generation_Cloud_Computing
24. [PDF] Supply chain analysis and material demand forecast in strategic ... https://rmis.jrc.ec.europa.eu/uploads/CRMs_for_Strategic_Technologies_and_Sectors_in_the_EU_2023.pdf
25. A Testing Laboratory of Optimization Techniques in Unreal ... https://arxiv.org/html/2604.02980v1
26. Profiling with Purpose: Performance Lessons from a Real ... https://dev.epicgames.com/community/learning/tutorials/qEzo/unreal-engine-profiling-with-purpose-performance-lessons-from-a-real-unreal-project
27. Unreal Engine Optimization Guide: Profiling Fundamentals https://www.intel.com/content/www/us/en/developer/articles/technical/unreal-engine-optimization-profiling-fundamentals.html
28. BCI-UE4-Plugin - Brain-Computer-Interface for Unreal ... https://forums.unrealengine.com/t/bci-ue4-plugin-brain-computer-interface-for-unreal-engine-4/68495
29. [PDF] Science and Practice of Pediatric Critical Care Medicine https://www.researchgate.net/profile/William-Novick/publication/227040219_Pediatric_Critical_Care_in_Developing_Countries/links/0046351a748b64c2ce000000/Pediatric-Critical-Care-in-Developing-Countries.pdf
30. COMPLETE SUPERCONDUCTIVE CULTURE.doc - Academia.edu https://www.academia.edu/38494725/COMPLETE_SUPERCONDUCTIVE_CULTURE_doc
31. (PDF) Alignment Faking - the Train -> Deploy Asymmetry: Through a ... https://www.researchgate.net/publication/397934316_Alignment_Faking_-_the_Train_-

_Deploy_Asymmetry_Through_a_Game-Theoretic_Lens_with_Bayesian-Stackelberg_Equilibria

32. [PDF] 北京邮电大学科技论文，专著（译著）统计 https://kyy.bupt.edu.cn/__local/D/6B/40/52CFE60232B89AD9E602D5CFCCB_5CD1F708_7D7F7B.pdf?e=.pdf
33. The Semantic Web – ISWC 2025 - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-032-09530-5.pdf
34. validate nested content of json schema https://stackoverflow.com/questions/67112862/validate-nested-content-of-json-schema
35. Techno Vision 2026 https://www.capgemini.com/wp-content/uploads/2026/02/Capgemini_TechnoVision2026.pdf
36. 333333 23135851162 the 13151942776 of 12997637966 ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/words-333333.txt
37. fNIRS-EEG BCIs for Motor Rehabilitation: A Review - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10740927/
38. [PDF] Broadband networks of the future | OECD https://www.oecd.org/content/dam/oecd/en/publications/reports/2022/07/broadband-networks-of-the-future_cb6dd626/755e2d0c-en.pdf
39. Islands of education: schooling, civil war and the Southern ... https://unesdoc.unesco.org/ark:/48223/pf0000143160
40. What Develops in Emotional Development? - Springer Nature https://link.springer.com/content/pdf/10.1007/978-1-4899-1939-7.pdf
41. Scientific psychiatry in Stalin's Soviet Union: The politics of modern ... https://search.proquest.com/openview/7708dfc0df6e540f5f2f61fc192c499c/1?pq-origsite=gscholar&cbl=18750&diss=y
42. C++ Nested JSON in Unreal Engine 4 https://stackoverflow.com/questions/30342465/c-nested-json-in-unreal-engine-4
43. The Cloud Pod | Weekly AI & Cloud News on AWS, Azure & GCP https://podcasts.apple.com/no/podcast/the-cloud-pod/id1447083316
44. clusters_dj_1_2_3_generalized.txt - Google Groups https://googlegroups.com/group/opencog-soc/attach/dcb51e9b0dfac6e4/clusters_dj_1_2_3_generalized.txt?part=0.1
45. [PDF] TechnoVision 2025 - Capgemini https://www.capgemini.com/wp-content/uploads/2025/02/TV25-Report_Web-version.pdf
46. The Three-Tier Governance Architecture that Changes ... https://www.linkedin.com/pulse/three-tier-governance-architecture-changes-everything-chris-hood-mnavf
47. Claude Code's 3-Tiered Memory Structure Simplifies File ... https://www.linkedin.com/posts/subham-kundu-2746b515b_%F0%9D%97%9B%F0%9D%97%BC%F0%9D%98%84-

%F0%9D%97%B1%F0%9D%97%BC%F0%9D%97%B2%F0%9D%98%80-%F0%9D%97%BA%F0%9D%97%B2%F0%9D%97%BA%F0%9D%97%BC%F0%9D%97%BF%F0%9D%98%86-%F0%9D%97%AE%F0%9D%97%B0%F0%9D%98%81%F0%9D%98%82%F0%9D%97%AE%F0%9D%97%B9-activity-7445848581237940225-E-79

48. (PDF) Blockchain Enabled Applications Understand the ... https://www.researchgate.net/publication/362689200_Blockchain_Enabled_Applications_Understand_the_Blockchain_Ecosystem_and_How_to_Make_it_Work_for_You
49. Yong-Ku Kim Editor - Anxiety Disorders - Springer Nature https://link.springer.com/content/pdf/10.1007/978-981-32-9705-0.pdf
50. 30th Annual Computational Neuroscience Meeting - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC8687879/
51. (PDF) Fear Level Classification Based on Emotional Dimensions ... https://www.researchgate.net/publication/332375836_Fear_Level_Classification_Based_on_Emotional_Dimensions_and_Machine_Learning_Techniques
52. A Reference Architecture for Governance of Cloud Native Applications https://arxiv.org/html/2302.11617v3
53. Tier-1 Gateway Configurations in NSX Federation https://techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0/advanced-network-management/administration-guide/managing-nsx-t-in-multiple-locations/nsx-t-federation/networking-topologies-in-nsx-federation/tier-1-in-federation.html
54. Deliverable D3.3 5G NORMA network architecture https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e5b5c2a614&appId=PPGMS
55. [PDF] VMware Government Services IL5 Tenant Configuration Guide ... https://www.vmware.com/docs/vmw-vgs-il-five-tenant-configuration-guide
56. Meet 3D-GPT: An Artificial Intelligence Framework for Instruction ... https://www.marktechpost.com/2023/10/28/meet-3d-gpt-an-artificial-intelligence-framework-for-instruction-driven-3d-modelling-that-makes-use-of-large-language-models-llms/
57. Tech Stories Topics - HackerNoon https://hackernoon.com/c/tech-stories/topics
58. (PDF) WEB - Academia.edu https://www.academia.edu/28317794/WEB
59. Accepted CMP Alerts 20230504 002531 | PDF - Scribd https://www.scribd.com/document/642774132/accepted-cmp-alerts-20230504-002531-xlsx
60. Encapsulated Functions:Fortifying Rust's FFI in Embedded ... https://sns.cs.princeton.edu/assets/papers/2023-encapfn-schuermann.pdf

61. Social Work Treatment Interlocking Theoretical Approaches (Turner ... https://www.scribd.com/document/697463538/Social-Work-Treatment-Interlocking-Theoretical-Approaches-Turner-Francis-Joseph-Ed-Z-lib-org
62. Symmetry, Volume 17, Issue 11 (November 2025) – 234 articles https://www.mdpi.com/2073-8994/17/11
63. Cybersecurity Preparedness 2025 Report | PDF | Security - Scribd https://www.scribd.com/document/839414992/Cybersecurity-Fraud-Risk-Report-2025
64. The Cloud Pod - Apple Podcasts https://podcasts.apple.com/sl/podcast/the-cloud-pod/id1447083316
65. The Construction of Spanish National Identity - jstor https://www.jstor.org/stable/pdf/j.ctvgc62dd.pdf
66. (PDF) Pediatric Intensive Care Unit Administration - Academia.edu https://www.academia.edu/17759314/Pediatric_Intensive_Care_Unit_Administration
67. [PDF] D2.4 Platform Architecture and Specifications - European Commission https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e5e648af25&appId=PPGMS
68. On-the-Fly Fusion of Remotely-Sensed Big Data Using an Elastic ... https://pmc.ncbi.nlm.nih.gov/articles/PMC8122984/
69. 2026 Technology Trends: Augmented Intelligence | PDF - Scribd https://www.scribd.com/document/984535165/Ten-Technology-Trends-2026
70. BCI-UE4-Plugin - Brain-Computer-Interface for Unreal ... https://forums.unrealengine.com/t/bci-ue4-plugin-brain-computer-interface-for-unreal-engine-4/68495?page=3
71. Trends and Innovations in Information Systems and Technologies https://link.springer.com/content/pdf/10.1007/978-3-030-45691-7.pdf
72. Search Results for “guide accessible submission” - CHI 2018 - ACM https://chi2018.acm.org/?s=guide+accessible+submission&submit=
73. HCI International 2022 – Late Breaking Papers - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-031-22131-6.pdf
74. Technological Innovation for Applied AI Systems - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-030-78288-7.pdf
75. 2022 Index IEEE Internet of Things Journal Vol. 9 https://ieeexplore.ieee.org/iel7/6488907/9973413/10004784.pdf
76. Parallel and Distributed Computing, Applications and Technologies https://www.scribd.com/document/781216307/Lecture-Notes-in-Electrical-Engineering-1112-Ji-Su-Park-Hiroyuki-Takizawa-Hong-Shen-James-J-Park-Parallel-and-Distributed-Computing-Applicat

77. Appl. Sci., Volume 15, Issue 4 (February-2 2025) – 573 articles - MDPI https://www.mdpi.com/2076-3417/15/4
78. Futuristic Trends in ARTIFICIAL INTELLIGENCE - Academia.edu https://www.academia.edu/144848365/Futuristic_Trends_in_ARTIFICIAL_INTELLIGENCE
79. Google 1-gram - CMU School of Computer Science https://www.cs.cmu.edu/~roni/11761/2017_fall_assignments/hw3_stats_google_1gram.txt
80. Advances in VLSI, Signal Processing and Wireless Communication https://link.springer.com/content/pdf/10.1007/978-981-96-1587-2.pdf
81. NLP-Spam Email Classifier using Naive Bayes - Kaggle https://www.kaggle.com/code/vrajm1209/nlp-spam-email-classifier-using-naive-bayes
82. Tier Classification System - Uptime Institute https://uptimeinstitute.com/tiers
83. LNCS 8522 - Human Interface and the Management of Information https://link.springer.com/content/pdf/10.1007/978-3-319-07863-2.pdf
84. (PDF) Design and implementation of a scalable and high-throughput ... https://www.researchgate.net/publication/386589496_Design_and_implementation_of_a_scalable_and_high-throughput_EEG_acquisition_and_analysis_system
85. 计算机视觉与模式识别2024_6_18 - arXiv每日学术速递 http://arxivdaily.com/thread/56542
86. Appl. Sci., Volume 15, Issue 17 (September-1 2025) – 625 articles https://www.mdpi.com/2076-3417/15/17
87. Leadership Styles and Time Management | PDF | Internet - Scribd https://www.scribd.com/document/451354867/words-333333-txt
88. words_SG_upto2020.txt - Zenodo https://zenodo.org/record/5516252/files/words_SG_upto2020.txt
89. Is there a way of making serde_json deserialize strictly? https://stackoverflow.com/questions/60489085/is-there-a-way-of-making-serde-json-deserialize-strictly
90. Australian Stock Margin Rates Overview | PDF - Scribd https://www.scribd.com/document/294089035/Current-Stock-Deal-Settings-4